Block 1 — Week 2

Common Data Models: OMOP/OHDSI in Depth

Essentials of Biomedical Informatics — Self-Paced Module (Asynchronous)

Carlos E. Figueroa Castro, MD. MS
Associate Professor, Infectious Diseases
Medical College of Wisconsin
Milwaukee, WI

Overview

This week moves from generic data handling (Week 1) into the specific data model used throughout the rest of the course: The Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM). Plan for roughly 3–4 hours.

Companion Files

- synthetic_omop.duckdb — shared teaching database (it will be used again in Weeks 3 and 4)
- Week2_OMOP_Exploration.sql — worked SQL walkthrough + exercise

This is a small, hand-built database using the real OMOP CDM table and column names, with a tiny 20-row vocabulary standing in for the full Athena vocabulary (https://athena.ohdsi.org/search-terms/start), which has millions of concepts. The query patterns are identical to what you'd use on a real OMOP database.

Learning Objectives

- Describe the structure of the OMOP CDM: tables, domains, relationships
- Explain why standardized vocabularies (SNOMED CT, RxNorm, LOINC, ICD) matter for pooling data across institutions
- Describe the Extract, Transform, Load (ETL) process that transforms raw source data into the CDM
- Identify the purpose of core Observational Health Data Sciences and Informatics (OHDSI) collaborative tools: Atlas (https://atlas-demo.ohdsi.org/#/home), Achilles (https://ohdsi.github.io/Achilles/), and HADES (https://ohdsi.github.io/Hades/)
- Explain what a federated research network is and why data governance matters in that context

Core Concepts

The OMOP CDM is a relational schema that any health dataset — an EHR export, a claims database, a registry — can be mapped into, so that the same analysis code works regardless of where the data originally came from. Each CDM table represents a domain of clinical fact: Person (who), Condition_Occurrence (diagnoses), Drug_Exposure (medications), Measurement (labs and vital signs), and dozens of others. Every clinical fact links back to a person_id, and every row's meaning is anchored not in free text but in a concept_id — a reference to a single, standardized vocabulary term.

This is the second core idea: standardized vocabularies. Real-world source data arrives in every imaginable local format — one system might record a diagnosis as free text, another as a local billing code, another as an ICD-10 code. Standard vocabularies like SNOMED CT (conditions/procedures), RxNorm (medications), and LOINC (labs) give every concept a single canonical identity, so that “type 2 diabetes” means the same thing whether it came from a hospital in Milwaukee or a clinic in Seattle.

Getting from messy source data to a clean CDM is the job of ETL: extracting records from the source system, transforming local codes into standard concept_ids (a process called vocabulary mapping), and loading the result into CDM-shaped tables. This is usually the most labor-intensive part of any real-world OMOP project — far more time is typically spent on ETL than on the downstream analysis.

Once data is in the CDM, the OHDSI community provides free, open-source tooling built directly on top of it:

- Atlas — a web application for building cohort definitions and running analyses without writing SQL directly
- Achilles — a data characterization tool that profiles a CDM database (counts, distributions, data quality checks)

- HADES — a collection of R packages for more advanced analytic methods (population-level estimation, patient-level prediction)

Finally, because CDM-formatted data looks the same everywhere, networks of institutions can run identical analysis code locally and share only aggregate results — never raw patient data. This is a federated research network. All of Us, N3C, and PCORnet (introduced in Week 1) all rely on this pattern, which is what lets them combine data from hundreds of sites while satisfying each site's data governance and privacy requirements.

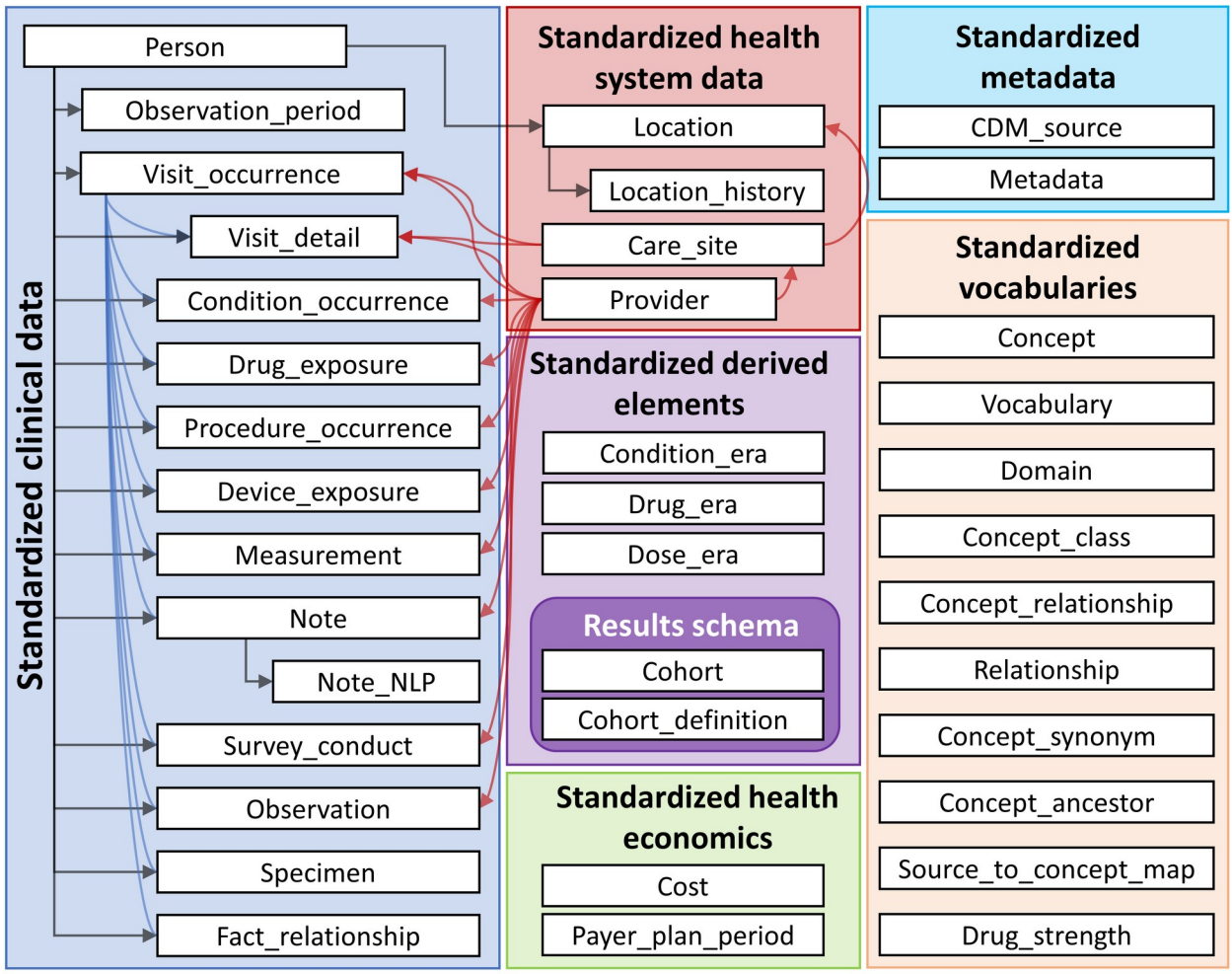


Figure 1: OMOP CMD overview (downloaded from https://ohdsi.github.io/TheBookOfOhdsi/CommonDataModel.html)

Setup

No new software beyond a SQL engine. Choose whichever path matches how you like to work:

Option A: DuckDB command-line (recommended, simplest)

- Install: https://duckdb.org/install — download the command line interface (CLI) for your operating system
- Download synthetic_omop.duckdb from Brightspace into a working folder
- Run: `duckdb synthetic_omop.duckdb` (opens an interactive SQL prompt)

Option B: From R

- `install.packages("duckdb")`
- `con <- DBI::dbConnect(duckdb::duckdb(), "synthetic_omop.duckdb")`

Option C: From Python

- `pip install duckdb`
- `import duckdb; con = duckdb.connect("synthetic_omop.duckdb")`

This Week's Deliverable

Open Week2_OMOP_Exploration.sql and work through Part 1 (Walkthrough) to see how the queries build on each other. Then complete Part 2 (Your Turn):

1. Write a query returning the top 10 medications by number of occurrences.
2. Write a query counting distinct patients (not occurrences) per condition.
3. Add a short comment explaining the difference between the two, and why it matters for real clinical data.

Submit the completed .sql file via the Week 2 dropbox in Brightspace, along with a comma separated value (CSV) or screenshot of your two query results.

Self-Check Quiz

Use this to check your own understanding before submitting. Answer the quiz in Brightspace.

1. What is the main purpose of the OMOP Common Data Model?

- a) To store data faster than a normal database
- b) To give datasets from different institutions a common structure so the same analysis code works on all of them
- c) To automatically de-identify patient data
- d) To replace the need for a SQL database entirely

2. Why does OMOP map local codes to standardized vocabularies like SNOMED CT or RxNorm?

- a) To make queries run faster
- b) So the same clinical concept means the same thing regardless of which institution the data came from
- c) Because it is legally required by HIPAA
- d) To reduce file size

3. What does the “T” in ETL refer to, and why is it often the most labor-intensive step?

- a) "Transfer" — moving files between servers
- b) "Transform" — mapping local source codes to standardized concept_ids
- c) "Test" — running unit tests on the pipeline
- d) "Tabulate" — counting rows

4. Which OHDSI tool is designed for data characterization (profiling a database's contents and quality)?

- a) Atlas
- b) Achilles
- c) HADES
- d) Athena

5. In a federated research network like N3C or PCORnet, what is typically shared between institutions?

- a) Raw patient-level records
- b) Aggregate results from analyses run locally on each site's own data
- c) Nothing is ever shared
- d) Only billing information

Suggested Readings

- Observational Health Data Sciences and Informatics. (2021, January 11). The Book of OHDSI. https://ohdsi.github.io/TheBookOfOhdsi/ (chapters 4–6)
- OMOP CDM documentation: https://ohdsi.github.io/CommonDataModel/

For further exploration: OHDSI's Eunomia package (https://ohdsi.github.io/Eunomia/) provides a larger synthetic OMOP database with the full standard vocabulary, useful once you want to go beyond this week's teaching-scale dataset.

Suggested Grading Rubric

A pass grade required 60/100 points

Criterion	Weight	What to look for
Walkthrough completion	25%	All Part 1 queries run without errors
Task 1: top medications query	25%	Correct join and aggregation logic
Task 2: distinct patients query	25%	Correct use of COUNT(DISTINCT ...)
Task 3: written explanation	25%	Correctly distinguishes occurrences vs. patients